

TinyML Smart Plug

Local socket-level protection for overload, socket heating, and series arc-fault events

User Manual Booklet

Prototype operating guide for setup, monitoring, data logging, CSV review, OTA handling, and TinyML training workflows.

Thesis Project Prototype

Firmware reference: v7.8.2-p-gen0-final

Suggested Canva Layout

Use a full-page product photo or clean device render. Keep the title large and plain. Add only one short purpose line under the project name.

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Note

This booklet is written for a general user first. Admin, OTA, Logger, CSV Editor, and TinyML Trainer pages are for the project developer or trained operator.

Suggested Canva Layout

Two-column table of contents. Add small icons for safety, Wi-Fi, relay, data, firmware, and training if desired.

About the Device

The TinyML Smart Plug is a smart plug prototype designed to add local protection at the socket level. It monitors voltage, current, estimated socket temperature, and waveform behavior to respond to overload, socket heating, and series arc-fault events.

The device can work with the PWA dashboard for monitoring, relay control, fault clearing, event history, data collection, CSV review, OTA controls, and TinyML development workflows.

Protection decisions are handled locally by the device firmware. The website improves visibility and control, but it is not required for the main protection loop.

What it helps detect

- Overload warning and sustained overload trip.
- Estimated socket heating warning and trip.
- Series arc-fault events using waveform features and TinyML inference.
- Abnormal voltage conditions, including under-voltage and over-voltage protection.

What it is not

- Not a certified replacement for circuit breakers, AFCIs, GFCIs, proper wiring, or electrical inspection.
- Not a waterproof or outdoor-rated product.
- Not a commercial production unit.

Warning

Use this prototype only in safe, supervised test conditions. Do not use it with damaged outlets, burned plugs, unstable wiring, wet locations, or loads beyond the intended test range.

Suggested Canva Layout

Left side: short description. Right side: simple block diagram from wall outlet to smart plug to protected appliance. Icons: plug, sensor, relay, display, Wi-Fi, cloud.

Key Features

Protection

- Local relay shutdown during critical fault conditions.
- Overload warning above 10.0 A.
- Sustained overload trip using averaged current behavior.
- Estimated socket heating warning at about 39.0°C and trip at about 40.0°C.
- Series arc-fault detection using waveform features and TinyML model output.

User operation

- PWA dashboard at `tinymml-smart-plug.web.app`.
- Relay ON/OFF control from the dashboard.
- Clear Fault command from the dashboard.
- Manual relay ON button for limited offline use.
- OLED display for local status and basic readings.

Developer and admin tools

- Firebase live data and event logging when Wi-Fi is available.
- Logger for controlled data collection sessions.
- CSV viewer, plotter, editor, arc marker tool, and segment labeling.
- OTA firmware controls and manual firmware download workflow.
- TinyML data preparation, model training, benchmarking, and exported firmware headers.

Note

Voltage protection is included because the firmware and manuscript define under-voltage and over-voltage states that can lock relay control during unsafe mains conditions.

Suggested Canva Layout

Use three clean feature groups: Protection, User Operation, Admin Tools. Use check icons or simple line icons. Avoid long paragraphs.

Important Safety Warnings

Warning

Always inspect the outlet, plug, and appliance before use. Do not use the device on a loose, damaged, burned, wet, or unstable outlet.

Warning

If the device trips, do not immediately turn the relay back ON. Inspect the connected appliance, socket, plug, wiring, and load condition first.

Warning

The prototype weighs around 750 g. A weak wall socket may not support it securely. Prevent sagging and falling before energizing the load.

- Do not use with**
- Loose wall sockets or protruding outlet adapters.
 - Burned, cracked, wet, corroded, or unstable outlets.
 - Damaged extension cords or overloaded power strips.
 - Appliances with visible plug, cord, or casing damage.
 - Loads that exceed the intended prototype test conditions.
- Stop use immediately if**
- There is smoke, burning smell, sparking, or unusual heat.
 - The enclosure, plug, socket, or cable becomes damaged.
 - The device repeatedly trips after clearing a fault.
 - The plug no longer fits tightly in the outlet.

Suggested Canva Layout

Use a red safety band with four warning cards. Add icons for heat, unstable socket, electric shock, and inspect before reset.

Physical Handling and Mounting

The TinyML Smart Plug is heavier than a normal plug-in device because of its relay, heatsink, enclosure, and protection circuitry. The current prototype weighs around 750 g. Because of this weight, the device may sag if it is plugged into a loose, protruding, or weak wall socket. The plug must remain tightly connected during operation.

Recommended mounting

1. Use a firm wall outlet that holds the plug tightly.
2. Position the device so it sits flat or nearly flush against the wall.
3. Support the device if needed using a safe stopper, backing, or spacer.
4. Keep the device stable before connecting the appliance load.

Using an extension wire or power strip

- Lay the smart plug flat on the floor or on a stable surface.
- Do not let the device hang from the outlet, cord, or power strip.
- Keep the power strip away from water, dust buildup, and foot traffic.

Warning

Do not use loose, damaged, burned, cracked, or unstable outlets. A loose connection can create heat and may make the socket heating condition worse.

Suggested Canva Layout

Show a correct/incorrect mounting comparison: flush supported wall mount, flat power strip setup, sagging unsupported outlet. Add callouts for plug fit, support surface, and protected socket.

System Components

Component	Purpose
Red main rocker switch	Powers the TinyML Smart Plug itself. This is different from the relay control for the appliance socket.
Protected outlet/socket	Output socket where the appliance or test load is connected. It only receives power when the relay output is ON.
Relay / solid-state relay path	Connects or disconnects power to the protected outlet during user control or protection trips.
ESP32-S3 controller	Runs Wi-Fi, local protection logic, sensor processing, TinyML inference, Firebase communication, and control handling.
Voltage sensor	Measures mains voltage for monitoring and abnormal voltage protection.
Current sensor and ADC	Measures load current and waveform behavior used by protection and TinyML features.
Thermistor	Measures internal/sensor temperature used to estimate socket heating behavior.
OLED display	Shows local status and limited live values such as voltage, current, estimated socket temperature, and apparent power.
Buzzer/chime	Gives audible feedback for setup, collection, faults, and command acknowledgment.
Manual relay ON button	Red circular non-tactile button beside the main switch. Used only to turn the relay ON during limited offline operation.

Note

This manual uses the user-facing term **relay output**. The technical prototype uses a relay/SSR switching path to control the protected socket.

Suggested Canva Layout

Use one device photo with numbered callouts. Suggested labels: Main Switch, Manual Relay ON Button, OLED, Protected Outlet, Plug Input, Enclosure, Vent/Heat Area if visible.

Quick Start

Basic setup

1. Plug the TinyML Smart Plug into a stable outlet.
2. Turn the red main switch ON.
3. Connect the device to a 2.4 GHz Wi-Fi network.
4. Open `tinymml-smart-plug.web.app` in Google Chrome.
5. Plug the appliance or test load into the protected outlet.
6. Turn the relay ON from the PWA rocker control.

Important behavior

- The relay is OFF by default after startup.
- Relay ON gives power to the appliance socket.
- Relay OFF removes power from the appliance socket.
- The red main switch powers the smart plug itself.

If Wi-Fi is unavailable

1. Keep the device plugged in and powered.
2. Connect the load first.
3. Hold the manual relay ON button for about 5 seconds.
4. Wait for the chime.
5. Release the button.

Warning

Use the manual relay ON button only when needed. It cannot turn the relay OFF. Use the PWA relay OFF command or remove power using the main switch if the load must be stopped.

Suggested Canva Layout

Make this a visual flow: Power, Wi-Fi, PWA, Plug Load, Relay ON. Add a small offline branch for the manual relay ON button.

Wi-Fi Setup

First-time setup

1. Turn the red main switch ON.
2. Open Wi-Fi settings on a phone or laptop.
3. Connect to the TinyML-Smart-Plug access point.
4. Wait for the captive portal to appear.
5. Select **Configure Wi-Fi**.
6. Choose a 2.4 GHz Wi-Fi network.
7. Enter the Wi-Fi password.
8. Wait for the device to chime and show Wi-Fi connected status on the OLED.

After setup

- Saved credentials are kept after reset.
- On the next startup, the device tries to reconnect automatically.
- The PWA can monitor and control the device when Wi-Fi and Firebase are reachable.

Warning

The device accepts 2.4 GHz Wi-Fi only. A 5 GHz-only network will not appear or will not connect.

Note

Firmware source sets the access point name to TinyML-Smart-Plug. If no saved Wi-Fi credentials exist, the boot AP waits about 15 seconds for a client. After a client connects and the portal opens, normal setup has about 45 seconds to complete.

Suggested Canva Layout

Use four screenshots: phone Wi-Fi list, captive portal, Configure Wi-Fi page, OLED Wi-Fi connected message. Add a 2.4 GHz badge.

Resetting the Device

Resetting restarts the smart plug and clears temporary runtime state. It does not erase saved Wi-Fi credentials.

How to reset

1. Turn the red main switch OFF.
2. Wait at least 10 seconds.
3. Turn the red main switch ON again.
4. Wait for startup and Wi-Fi reconnect.

Reset clears

- RAM state.
- Temporary buffers.
- Runtime timers.
- Active fault latch state after full restart.

Reset does not clear

- Saved Wi-Fi credentials.
- Firmware version.
- Cloud history already uploaded to Firebase.
- CSV files already downloaded or saved in the PWA workflow.

Note

The 10-second wait helps the device fully power down and gives capacitors time to discharge before restarting.

Warning

If the device tripped because of a fault, inspect the load, plug, socket, and wiring before resetting or turning the relay back ON.

Suggested Canva Layout

Use a simple OFF, wait 10 seconds, ON diagram. Add a small note that Wi-Fi credentials remain saved.

Offline Protection

The PWA is important for monitoring and control, but the protection loop is local. The device can still respond to overload, socket heating, abnormal voltage, and arc-fault conditions even when the website is not open.

When Wi-Fi is unavailable

- Local protection remains active.
- The relay can still shut OFF during critical fault conditions.
- The OLED shows limited live information.
- The buzzer can still give local feedback.

Limitations without Wi-Fi

- PWA relay control is unavailable.
- Firebase live data and event uploads may be delayed or missing.
- Data collection sessions cannot be fully managed from the website.
- CSV viewing, editing, and downloading are not available until the PWA has data.

Note

The OLED can show limited live values such as voltage, current, estimated socket temperature, and apparent power.

Suggested Canva Layout

Split the page into Online and Offline columns. Show protection logic inside the device, not inside the website.

Using the PWA

Open the dashboard

1. Open Google Chrome.
2. Go to `tinymml-smart-plug.web.app`.
3. Allow the page to load the dashboard.
4. Install the PWA if Chrome prompts installation.

For iPhone or iPad

- Open the site in Safari.
- Use **Share** then **Add to Home Screen**.
- Open the installed icon from the Home Screen.

Main dashboard areas confirmed in code

- System Overview.
- Relay rocker control.
- Refresh and Clear Fault controls.
- Status cards for load family, protection, fault timing, firmware, and last event.
- Main readings for voltage, current, sensor temperature, estimated socket temperature, expected normal temperature, and apparent power.
- Waveform disturbance and conduction intermittence feature cards.
- History section with date range, download, and clear actions.
- Admin-only Logger, Plot Viewer, CSV tools, OTA controls, and TinyML-related workflow support.

Suggested Canva Layout

Use a dashboard screenshot with numbered labels. Keep the PWA URL visible as a small footer callout.

Main PWA Interface

Area	What the user sees or does
System Overview	Shows device state, relay state, fault status, and quick controls.
Relay rocker	Turns the protected appliance socket ON or OFF when controls are available.
Clear Fault	Sends a fault clear command to the device. Use only after inspection.
Main Readings	Shows voltage, current, sensor temperature, estimated socket temperature, expected normal temperature, and apparent power.
Status cards	Show load family, protection state, fault timing, firmware information, and last event.
Feature cards	Show waveform and conduction features used for analysis and TinyML behavior.
History	Shows alerts/events with range filters such as Today, Yesterday, Last 7 days, and Last 30 days.
Menu	Contains notification mode, User/Admin mode switch, Change WiFi, OTA guide, and related admin controls.

Note

The PWA source uses a top-right menu button. This manual describes it as the three-line menu for clarity.

Suggested Canva Layout

Use one full-width PWA screenshot. Add callouts: Relay, Clear Fault, Readings, Status Cards, History, Menu.

Relay Control

The relay rocker at the top of the PWA controls the relay output. This controls electricity to the protected appliance socket.

Relay states

- **Relay ON:** the connected appliance receives power.

- **Relay OFF:** the connected appliance is turned off.

- **Red main switch ON:** the smart plug itself is powered.

- **Red main switch OFF:** the whole smart plug is powered down.

How to turn the appliance ON

1. Confirm the outlet and device are stable.
2. Plug the appliance into the protected outlet.
3. Open the PWA.
4. Move the relay rocker to ON.
5. Confirm that the load receives power.

Note

The relay is OFF by default after startup. This prevents the connected appliance from energizing automatically.

Warning

If the PWA blocks Relay ON, do not force operation. The device may be offline, disconnected, in startup, in OTA/config mode, or in a fault condition.

Suggested Canva Layout

Show a simple distinction: main switch powers device; relay rocker powers protected socket. Use two separate arrows.

Fault Warnings and Critical Trips

Fault behavior may be a warning or a critical trip.

Warning

- The device detects a condition that needs attention.
- The relay may remain ON while the condition is watched.
- The PWA and OLED may show warning status.

Critical trip

- The relay turns OFF as a system response.
- The connected appliance loses power.
- The user cannot normally turn the relay back ON until the fault is cleared.
- The device may chime, update Firebase/PWA status, and log an event if Wi-Fi is available.

Fault types used by firmware

- Overload warning.
- Sustained overload trip.
- Heating warning and heating trip.
- Arc-fault trip.
- Under-voltage and over-voltage lockout.
- Unplugged, disconnected, startup, OTA, Wi-Fi, or safe-mode states may also limit controls.

Note

The firmware gives priority to critical states such as heating trip, arc, sustained overload, and voltage lockout when deciding relay behavior.

Suggested Canva Layout

Use a two-level status diagram: Warning means watch and inspect; Critical Trip means relay OFF and inspect before clearing.

Clearing Faults

Faults can be cleared in two ways.

Option 1: PWA Clear Fault

- 1. Open the PWA dashboard.
- 2. Inspect the displayed fault condition.
- 3. Inspect the appliance, plug, socket, wiring, and load condition.
- 4. Click **Clear Fault**.
- 5. Wait for the device chime.
- 6. Turn the relay ON only if the setup is safe.

Option 2: Device reset

- 1. Turn the red main switch OFF.
- 2. Wait at least 10 seconds.
- 3. Turn the red main switch ON.
- 4. Wait for startup.
- 5. Turn the relay ON only after inspection.

Note

The firmware acknowledges the Clear Fault command with a local chime.

Warning

Clearing a fault means the user acknowledges that a problem occurred. Do not clear a fault just to continue operation. Inspect and address the connected appliance, socket, plug, wiring, or load condition first.

Suggested Canva Layout

Use an inspection checklist graphic before the Clear Fault button. Add a red “Inspect First” label.

Manual Relay ON Button

The red circular non-tactile button beside the main rocker switch is the manual relay ON button.

It is intended for limited use when Wi-Fi or the PWA is unavailable. It can turn the relay ON, but it cannot turn the relay OFF.

How to use

- 1. Connect the load to the protected outlet first.
- 2. Make sure the device is stable and powered.
- 3. Hold the manual relay ON button for about 5 seconds.
- 4. If a load is detected, the relay turns ON and the device chimes.
- 5. Release the button after the chime.

If no load is detected

- Holding the button for 5 seconds will not turn the relay ON.
- Confirm that a load is connected.
- Confirm that the appliance switch is in a state that draws detectable current.

To turn the relay OFF

- Use the PWA relay OFF command when available.
- A critical fault can also force the relay OFF.
- Turn the red main switch OFF if the device must be fully powered down.

Warning

This button is directly connected to the relay driver path and can bypass software restrictions. It must not be stuck, blocked, accidentally held, taped down, or compromised by any object.

Note

Firmware confirms a 5-second current-based physical relay ON confirmation. The manuscript and hardware notes describe this as the red manual relay ON button beside the main switch.

Suggested Canva Layout

Use a close-up photo of the red circular button with a warning callout. Add a small timing bar: hold 5 seconds, wait for chime, release.

Changing Wi-Fi

Use Change Wi-Fi when the router changes, credentials are wrong, or the device must connect to a different 2.4 GHz network.

From the PWA

1. Open the PWA dashboard.
2. Click the three-line menu at the top-right.
3. Click **Change WiFi**.
4. Wait for the device access point to open.
5. Open phone or laptop Wi-Fi settings.
6. Connect to the TinyML-Smart-Plug access point.
7. Follow the same Wi-Fi setup process.

If the device is not connected

- Reset the device using the red main switch.
- Watch for the access point during startup.
- Connect and configure the new 2.4 GHz Wi-Fi network.

Note

Changing Wi-Fi does not change the firmware or delete historical cloud records.

Note

For Change Wi-Fi, firmware opens a manual AP wait window of about 45 seconds and allows a longer manual portal session of about 4 minutes once the portal is active.

Suggested Canva Layout

Use three screenshots: menu open, Change WiFi button, Wi-Fi captive portal. Add a small 2.4 GHz-only callout.

Admin Mode

Admin mode opens development and maintenance tools. It is intended for the project developer or trained operator.

How to open Admin mode

1. Open the PWA dashboard.
2. Click the top-right menu.
3. Find the **Mode** switch.
4. Change from **User** to **Admin**.

Admin-only areas confirmed in code

- OTA controls.
- Logger controls.
- Original and Processed session tabs.
- CSV upload.
- Plot Viewer and CSV editor.
- Save and download processed CSV tools.

Admin controls in the menu

- Notification mode: Silent or Alert.
- Mode: User or Admin.
- Change WiFi.
- Revert Firmware.
- OTA Guide.

Warning

Admin tools can change data collection, firmware update, and processed dataset outputs. Use them only when the test setup is stable and the operator understands the workflow.

Suggested Canva Layout

Use a menu screenshot with the User/Admin switch highlighted. Add a lock or wrench icon for developer tools.

OTA Firmware Update

OTA controls are for the developer. Normal users should not update firmware unless instructed.

Current status

- HTTPS OTA is present in the firmware and PWA.
- The user notes say the HTTPS OTA path is currently not working properly.
- Only the developer has access to firmware version releases.

When to check updates

- Check when the firmware version shown on the website differs from the version shown in the menu or firmware information area.
- Current firmware reference in source: `v7.8.2-p-gen0-final`.

Manual update workflow

1. In Admin OTA, click the green download arrow to download the `.bin` firmware file.
2. Open the device AP using **Change WiFi**, or reset the device if it is not connected.
3. Connect to the TinyML Smart Plug AP.
4. Turn off mobile data.
5. Exit the captive portal window.
6. Open a real browser window.
7. Go to `192.168.4.1`.
8. Click **Update Firmware**.
9. Upload the downloaded `.bin` file.
10. Wait for upload, flashing, and automatic restart.

Warning

Do not remove power during firmware upload or flashing. Use a stable outlet and wait for the device to restart.

Note

The main firmware and PWA support HTTPS OTA through the published version and firmware URL fields. The repository also contains a browser upload OTA utility route at `/update`. For the manual local upload workflow, leave the captive portal and use a real browser at `192.168.4.1`.

Suggested Canva Layout

Use a firmware update timeline: Download BIN, Open AP, Real Browser, Upload, Flash, Restart. Add a warning icon at the flashing step.

Data Collection / Logger

The Logger records controlled sessions for analysis and TinyML dataset preparation. This section is for Admin users.

Before collecting

- Use a stable outlet and safe mounting.
- Confirm the load type and load name.
- Decide the phase: Start, Steady, or Arc.
- Keep the setup supervised.

Logger fields confirmed in PWA

- Duration in seconds.
- Load type or family.
- Device/load name.
- Trial number.
- Division tag: start, steady, or arc.
- Notes.
- Trusted normal switch.

How to collect

1. Open Admin mode.
2. Go to the Logger section.
3. Enter collection duration in seconds.
4. Select load type/family.
5. Enter load name and trial number.
6. Select start, steady, or arc phase.
7. Click or switch **Collect**.
8. Wait for the device chime.
9. Wait until the Collect rocker turns off automatically.
10. Wait for buffered upload to finish.

Note

The CSV time column may continue counting temporarily after collection ends because buffered samples are still uploading. When counting stops, the CSV is ready for viewing or downloading.

Suggested Canva Layout

Use a form screenshot with callouts for Duration, Load Type, Trial, Phase, Collect switch, and session tabs.

CSV Plotter and Editor

The Plot Viewer opens logged or uploaded CSV sessions for review, graphing, labeling, and download.

Opening CSV data

- Use Original sessions to view raw logger uploads.
- Use Processed sessions to view saved edited outputs.
- Use Upload CSV to open a local CSV file.
- If cloud saving fails after upload, the file can still open in the local viewer.

Graphing and viewing

- The viewer plots time-based CSV columns.
- Series can be searched, shown, hidden, set to default, set to all, or set to none.
- Visible series can be assigned to Y1 or Y2 axis.
- Controls include normalize, smooth, curve style, follow, zoom, play/pause, speed, and scrubber.

Data columns and features

- Voltage, current, temperature, and apparent power can be viewed when present.
- Waveform and feature columns can be plotted when present.
- The PWA recognizes arc labels using `label_arc`.
- CSV metadata can include family, device/load name, trial, notes, trusted normal, and division tag.

Saving

- Download CSV saves the current edited CSV.
- Save stores processed CSV output to the processed session area when Firebase saving is available.
- Download All can download sessions in order.

Note

The exact plotted series depends on the CSV headers available in the selected session. Missing columns do not appear as graphable series.

Suggested Canva Layout

Use a large plot screenshot. Add labels for Series panel, Zoom, Smooth, Normalize, Play/Pause, Save, and Download CSV.

Arc Markers and Section Labels

The CSV editor supports arc marker editing and waveform section labeling for TinyML dataset preparation.

Arc marker editing

- Enable Show arcs to display arc labels.
- Use the arc rows field to enter row numbers.
- Single rows and ranges are supported, such as 10–14,25.
- Click Add to mark rows as arc.
- Click Remove to clear arc markers.
- Edited arc labels are stored in `label_arc`.

Section labels

- Use Label to select Start, Steady, or Arc.
- Use Start and End row fields to define a section.
- Set Start and Set End can use the current viewer position.
- Add creates a labeled segment.
- Remove deletes a selected segment.
- Split downloads can create separate processed CSV files from labeled sections.

How phases are used

- Start rows are used for load context training.
- Steady and Arc rows are used for arc training data.
- Arc rows should represent the event period clearly.

Warning

Incorrect arc labels can reduce model quality. Review the waveform and event timing before saving processed CSV files.

Suggested Canva Layout

Show the editor panel with the arc row input and segment list. Add callouts for Label, Start row, End row, Add, Remove, and Download Split.

TinyML Trainer Workflow

The TinyML Trainer workflow is for the developer. It converts collected CSV logs into training datasets, model reports, and firmware header files.

Folder workflow confirmed in code

- Raw CSV files go in `tinymml/data/{gen}/raw`.
- Generation folders use names such as `gen0`, `gen1`, and later generations.
- Prepared outputs are written to `tinymml/data/{gen}/arc_training.csv` and `tinymml/data/{gen}/load_context.csv`.
- Reports are written to `tinymml/benchmark/{gen}`.
- Active firmware headers are written to `tinymml/model`.
- Archive copies are kept in generation-specific archive folders.

Preparation step

- Run `prepare_data.py` with the target generation.
- The script reads raw CSV files from the generation raw folder.
- It cleans and normalizes CSV columns.
- Start rows are kept for context training.
- Steady and arc rows are kept for arc training.
- The output files are `arc_training.csv` and `load_context.csv`.

Trainer GUI

- `tinymml/tinymml_trainer.py` provides a GUI workflow.
- The GUI shows the generation workflow and can help run preparation, training, and export tasks.

Suggested Canva Layout

Use a folder pipeline diagram: Raw CSVs, Prepare Data, Arc Dataset, Context Dataset, Train, Benchmark, Export Headers, Firmware.

TinyML Training and Export

Arc model training

- `train_rf.py` trains the Random Forest arc model.
- `train_et.py` trains the Extra Trees arc model.
- `train_duel.py` can compare RF and ET workflows.
- `train_feature_subsets.py` supports feature subset sweeps.
- Reports and feature summaries are saved in the benchmark folder.

Context model training

- `train_context.py` trains the load family context model.
- The context model classifies six load families.
- Context output is used as extra input for arc inference.

Firmware deployment

- Exported `.h` headers are generated in `tinym1/model`.
- Firmware includes these generated headers during build.
- Rebuild and flash the firmware after updating active model headers.
- Keep the generation manifest and benchmark report with the exported model.

Current active exported inputs

- Current RF header uses 13 inputs: 6 waveform/base features and 7 context inputs.
- Current context header uses 6 input features.

Note

The active exported RF header in `tinym1/model` uses 13 arc model inputs. An archived gen0 RF header used 12 inputs, but the current active header and manuscript workflow use the 13-input deployment structure.

Suggested Canva Layout

Use two side-by-side boxes: Arc Model and Context Model. Add arrows from Context Model to Arc Model.

Current Model Feature Reference

Model	Current exported input order confirmed in active headers
Arc RF base features	max_low_current_run_ms, zero_dwell_ratio, low_current_ratio, abs_irms_zscore_vs_baseline, midband_residual_ratio, spectral_flux_midhf.
Arc RF context inputs	ctx_family_resistive_linear, ctx_family_inductive_motor, ctx_family_rectifier_smps, ctx_family_phase_angle_controlled, ctx_family_brush_universal_motor, ctx_family_other_mixed, context_family_confidence.
Context model features	delta_irms_abs, halfcycle_asymmetry, midband_residual_ratio, thd_i, spectral_flux_midhf, hf_energy_delta.
Load family classes	resistive_linear, inductive_motor, rectifier_smps, phase_angle_controlled, brush_universal_motor, other_mixed.

Note

These lists are from the current exported headers in `tinym1/model`. If new headers are exported, update this page before printing the manual.

Suggested Canva Layout

Use this as a technical appendix-style spread. Keep it clean and table-based for developer reference.

Maintenance Guidelines

Before each use

- Check that the outlet is firm and undamaged.
- Check that the smart plug sits securely.
- Check the protected outlet and appliance plug.
- Keep the device dry and clear of clutter.

After a trip

- Disconnect or turn off the appliance.
- Inspect for heat, smell, discoloration, looseness, or visible damage.
- Do not continue repeated tests without identifying the cause.

Storage

- Store in a dry indoor location.
- Do not place heavy items on the enclosure, plug, or button.
- Keep dust away from openings and connectors.

Cleaning

- Turn the device OFF and unplug it.
- Use a dry cloth only.
- Do not spray liquid cleaner into the enclosure or socket.

Warning

Do not open the enclosure while the device is connected to mains power. Internal parts may carry dangerous voltage.

Suggested Canva Layout

Use a four-card maintenance layout: Inspect, Clean, Store, After Fault.

Maintenance Schedule

When	Action	Purpose
Before every use	Inspect outlet, plug fit, enclosure, protected socket, and appliance cord.	Prevent loose contact, overheating, and unsafe operation.
Before every use	Confirm the device is supported and not sagging.	Keep the plug tightly connected.
During use	Monitor OLED/PWA status and listen for chimes or alerts.	Catch warnings before continuing operation.
After any trip	Inspect load, socket, plug, wiring, and device temperature before clearing.	Avoid restarting into an unresolved fault.
Weekly during testing	Check for dust, discoloration, looseness, cracks, or button obstruction.	Maintain safe mechanical condition.
Before data collection	Confirm load label, phase, trial number, and logger settings.	Keep datasets organized and useful.
Before firmware update	Confirm firmware file, stable power, and correct device AP/browser workflow.	Reduce update failure risk.

Suggested Canva Layout

Use a clean schedule table. Color-code everyday, after-fault, and developer-only tasks.

Troubleshooting Guide

Problem	Possible cause	What to do
Device does not power on	Main switch OFF, outlet has no power, loose plug, or damaged outlet.	Turn main switch ON. Try a known good outlet. Do not use loose or damaged outlets.
Appliance does not turn on	Relay is OFF by default, PWA controls are locked, fault is active, or load is not connected correctly.	Open PWA and turn relay ON if safe. Check fault status. Inspect before clearing any fault.
Cannot connect to Wi-Fi	Using 5 GHz network, wrong password, AP window closed, or weak signal.	Use 2.4 GHz Wi-Fi. Reset and try setup again. Confirm AP name and password.
Captive portal does not open	Browser or phone did not launch portal automatically.	Open browser and go to 192.168.4.1 while connected to the device AP. Turn off mobile data if needed.
PWA shows stale or offline	Device not connected to Wi-Fi/Firebase, browser cache, or network issue.	Refresh the PWA. Check Wi-Fi. Reset the device if needed.
Relay ON is blocked	Fault latch, web controls locked, startup, OTA, config portal, disconnected, or unsafe state.	Read the status message. Inspect the setup. Clear fault only after inspection.
Fault returns after clearing	The appliance, outlet, wiring, or load condition may still be unsafe.	Stop using the setup. Inspect and correct the cause before trying again.
Manual relay ON button does not work	No detectable load, not held long enough, fault condition, or hardware issue.	Connect the load first, hold about 5 seconds, wait for chime. Do not force or jam the button.
Logger stops but CSV still counts	Buffered samples are still uploading.	Wait until counting stops before viewing or downloading the CSV.
Firmware update page not found	Captive portal window, AP not open, mobile data active, or local update route unavailable.	Use a real browser at 192.168.4.1, disable mobile data, and verify that the final firmware supports local Update Firmware.

Suggested Canva Layout

Use this as a full spread. Keep row height generous so Canva can place icons beside each problem.

Safety and Handling Checklist

Before powering ON

- Outlet is firm, dry, and undamaged.
- Device plug fits tightly.
- Smart plug is supported and not sagging.
- Appliance cord and plug are not damaged.
- Protected outlet is clear and dry.
- Manual relay ON button is not blocked or pressed.

Before turning relay ON

- Correct load is connected.
- Load is within intended test conditions.
- No active fault is displayed.
- No unusual heat, smell, smoke, or sound is present.

After a warning or trip

- Turn off or disconnect the load.
- Inspect the socket, plug, wiring, and appliance.
- Let hot parts cool before touching nearby surfaces.
- Clear the fault only after inspection.
- Do not repeat operation if the same fault returns.

For admin work

- Label the load, trial, and phase correctly.
- Keep raw and processed CSV files organized by generation.
- Do not update firmware on unstable power.
- Do not train or deploy models from poorly labeled data.

Suggested Canva Layout

Use checkboxes in Canva. This spread can also be printed as a lab checklist.

One-Page Quick Start Guide

1. Mount safely

- 1. Use a firm, undamaged outlet.
- 2. Keep the device flat, flush, or supported.
- 3. Do not let it hang from an extension cord or power strip.

2. Power the smart plug

- 1. Plug in the device.
- 2. Turn the red main switch ON.
- 3. Wait for startup.

3. Connect Wi-Fi

- 1. Connect to the TinyML-Smart-Plug AP.
- 2. Open Configure Wi-Fi.
- 3. Choose 2.4 GHz Wi-Fi.
- 4. Enter credentials.
- 5. Wait for chime and Wi-Fi connected status.

4. Open dashboard

- 1. Open Chrome.
- 2. Go to `tinymml-smart-plug.web.app`.
- 3. Install the PWA if prompted.

5. Power the appliance

- 1. Plug the appliance into the protected outlet.
- 2. Use the PWA relay rocker.
- 3. Relay ON powers the appliance.
- 4. Relay OFF turns the appliance off.

6. If a fault appears

- 1. Do not turn the relay back ON immediately.
- 2. Inspect the load, socket, plug, and wiring.
- 3. Click Clear Fault only after inspection.

Offline relay ON

- 1. Connect the load first.
- 2. Hold the red circular manual relay ON button for about 5 seconds.
- 3. Release after the chime.

Warning

The manual relay ON button cannot turn the relay OFF. It must never be stuck, blocked, or accidentally held.

Note

Reset: red switch OFF, wait 10 seconds, red switch ON. Wi-Fi credentials remain saved.

Suggested Canva Layout

This is the printable quick-start insert. Use large numbered blocks and minimal text.

Screenshots and Photos to Capture

Image	Use in booklet
Front photo of the TinyML Smart Plug	Cover page and About page.
Device plugged into a stable wall outlet	Physical handling and mounting.
Incorrect sagging setup, staged safely with no power	Safety warning comparison.
Device on a flat power strip or extension setup	Alternative safe mounting.
Close-up of red main switch	Basic use and reset pages.
Close-up of manual relay ON button	Manual button instructions and warning.
OLED startup and Wi-Fi connected screen	Wi-Fi setup and offline protection pages.
Phone/laptop Wi-Fi AP list	Wi-Fi setup.
Captive portal Configure Wi-Fi screen	Wi-Fi setup and Change Wi-Fi.
PWA dashboard main page	Using the PWA and relay control.
Fault state and Clear Fault button	Fault behavior and clearing faults.
Top-right menu with User/Admin switch	Admin mode and Change Wi-Fi.
OTA admin section with green download arrow	Firmware update workflow.
Logger section	Data collection instructions.
CSV Plot Viewer with graph and series panel	CSV plotter/editor page.
Arc marker and segment editor controls	Arc labels and section labels page.
TinyML trainer folder or GUI screenshot	TinyML trainer workflow page.

Suggested Canva Layout

Use this as a production checklist while collecting Canva assets. Keep filenames consistent with the page numbers.

Source Confirmation Notes

Item	Source-based decision used in this manual
Wi-Fi AP name	Firmware constant uses TinyML-Smart-Plug; this manual uses that exact SSID.
AP timeout timing	Firmware uses about 15 seconds for boot AP wait without a client, about 45 seconds for normal portal setup after a client connects, about 45 seconds for manual AP wait, and about 4 minutes for manual portal setup.
OLED wording after Wi-Fi connect	Notification source draws WIFI and CONNECTED; this manual uses Wi-Fi connected status.
Manual OTA path	Main firmware supports HTTPS OTA through Firebase/PWA fields. Repository utility code includes a browser upload OTA route at /update; the manual local workflow is kept as developer-only.
Firmware release access	The manual states that firmware releases are developer-controlled, matching the user's operating instruction.
Relay hardware label	The manual uses the user-facing phrase relay output and protected socket.
Manual relay ON button behavior	Firmware confirms 5-second current-based physical relay ON confirmation; user hardware notes identify the red circular button.
Current active model input count	Active RF header uses 13 inputs. Archived gen0 RF header used 12, but the active deployment uses 13.
Firmware version	Source shows v7.8.2-p-gen0-final; this is printed as the firmware reference.
PWA menu wording	PWA source contains a top-right menu with User/Admin mode, Change WiFi, OTA, and notification controls.
Voltage protection wording	Firmware and manuscript include under-voltage and over-voltage states; this manual includes them as control-locking protection states.

Suggested Canva Layout

Use this page as an appendix or internal source note. It explains why the manual uses the code-confirmed wording where earlier notes differed.

Contact and Support

For help

- Contact the thesis project developers or adviser.
- Provide the firmware version shown in the PWA.
- Describe the load used, outlet condition, and exact fault shown.
- Include screenshots of the PWA status and history entry if available.

Before requesting support

- Stop using the device if there is smoke, smell, heat, sparking, or visible damage.
- Do not clear repeated faults without inspection.
- Do not open the enclosure while connected to mains power.

Support information to fill in

Project team: [FILL IN]
Adviser: [FILL IN]
Email: [FILL IN]
Phone: [FILL IN]
Institution: [FILL IN]
Repo or release link: [FILL IN]

Suggested Canva Layout

Use a clean final page with contact fields, QR code to the PWA or repository if allowed, and a small product photo.

Glossary

Term	Meaning
Arc fault	An unsafe sparking or intermittent electrical condition that can occur through a damaged or loose connection.
CSV	Comma-separated values file used for logged sensor data and TinyML dataset preparation.
Estimated socket temperature	Firmware estimate of socket heating behavior based on sensor readings and model logic.
Firebase	Cloud service used by the project for live data, controls, event history, and logger uploads.
Firmware	Software running inside the ESP32-S3 smart plug controller.
Load	The appliance or test device plugged into the protected outlet.
OTA	Over-the-air firmware update process.
PWA	Progressive Web App. The dashboard can be opened in a browser and installed like an app.
Relay	Output switching path that energizes or disconnects the protected socket.
Sustained overload	Overload condition that remains long enough for the firmware to trip the relay.
TinyML	Machine learning model small enough to run locally on the microcontroller.
Trip	Protective action where the relay turns OFF because of a critical condition.

Suggested Canva Layout

Use as a final reference page. Add small icons only if space allows.